

SOLAR WEATHER

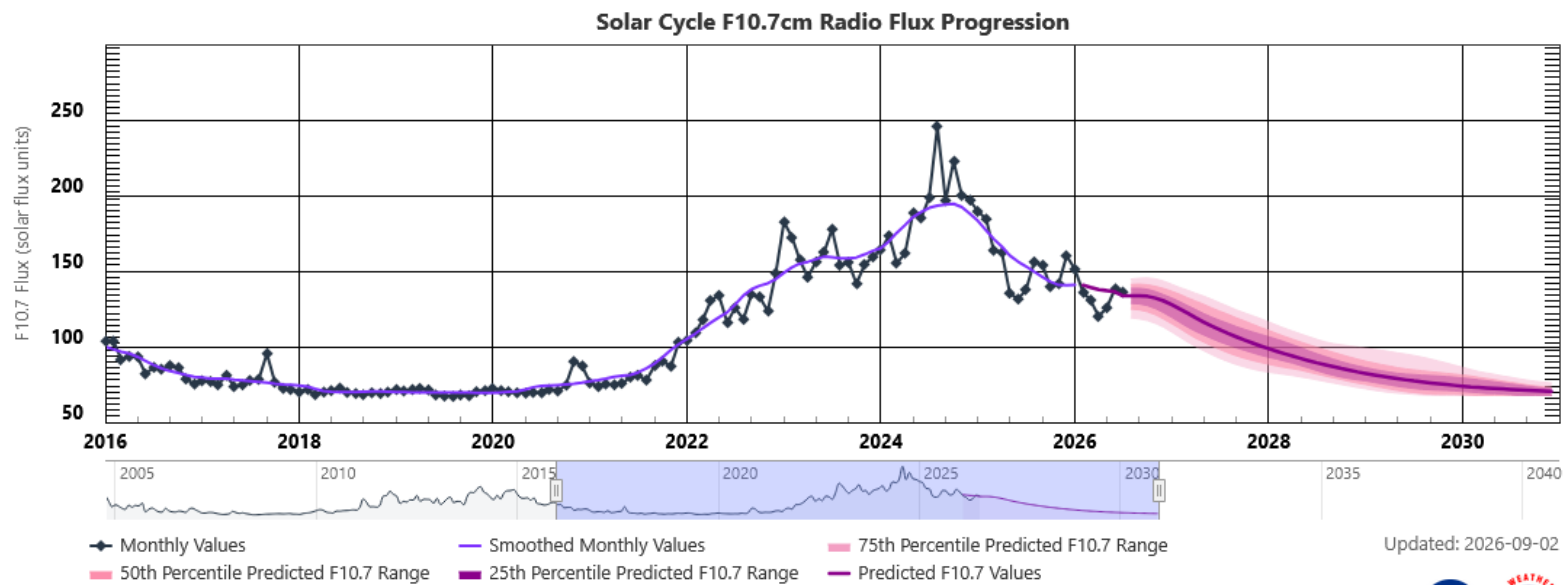
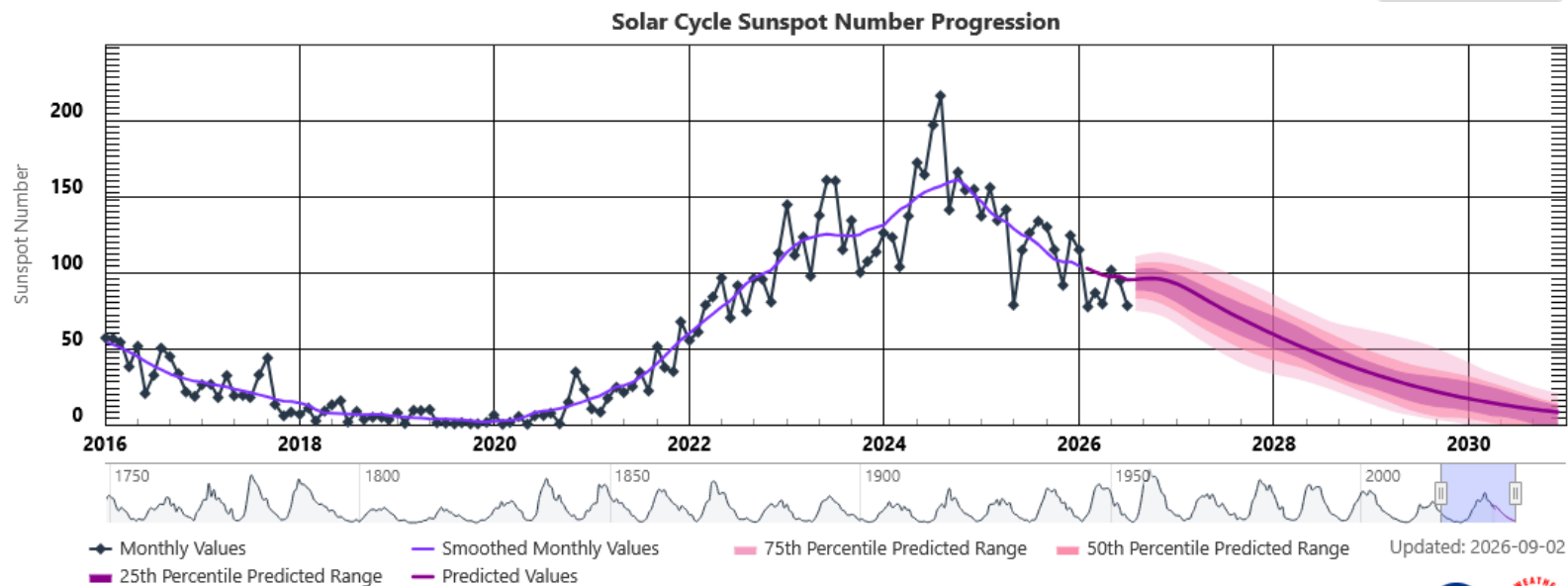
1 SEP 2026

Lewis Thompson
W5IFQ



Taken by Marianne
Bergli on August 28,
2026 @ Kvaløya,
Tromsø, Norway

Taken
October



SolarHam

Indices: (9/1 @ 00:35 UTC)

SFI 107

—

SSN 81

Geomagnetic Forecast

Sept 1

Sept 2

Sept 3

3 (G0)

2 (G0)

1-2 (G0)

Max Kp

M-Lat 05% M-Lat 01% M-Lat 05%

H-Lat 25% H-Lat 15% H-Lat 15%

Probabilities

[Detailed Forecast](#)

Solar Flare Probabilities

C-Flare: 65%

M-Flare: 10%

X-Flare: 01%

Proton: 01%

[Probability Details](#)

Updated @ 20:00 UTC (August 31)

Visible Sunspot Regions

AR 4523

A

N10E77

—

AR 4522

B

N13W15

—

AR 4521

A

N09E67

—

AR 4520

B

S12W07

▲

AR 4518

B

N07W24

▼

Updated @ 00:45 UTC (September 1)

[Sunspot Summary >](#)

[SRS](#)

[Data Archive](#)



Solar Flare Detection

Protons & Electrons

SWPC Display >

Current X-Ray Flux

B6.3

13:30 Sep 1

6-Hour Max

B9.4

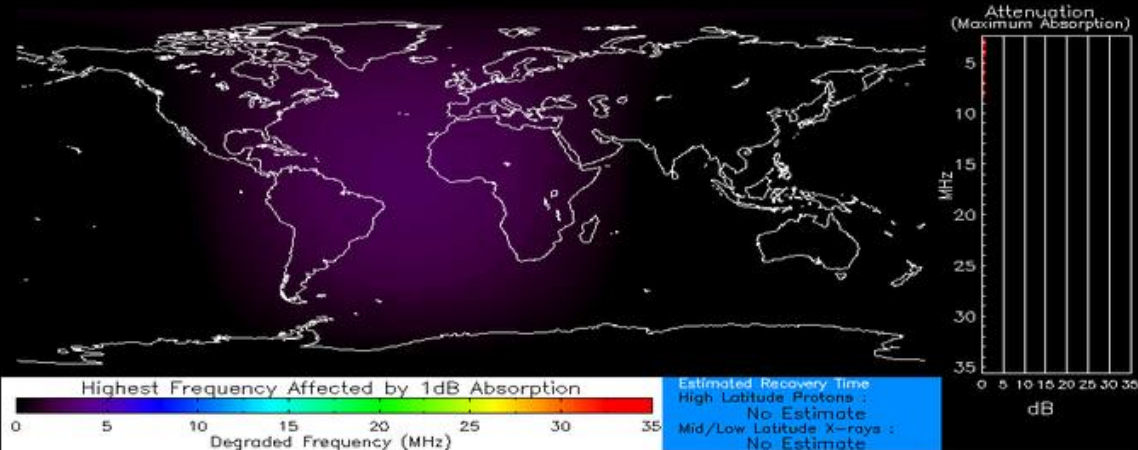
12:38 Sep 1

24-Hour Max

C2.6

15:16 Aug 31

X-Ray Data



Normal X-ray Background
Product Valid At : 2026-09-01 13:39 UTC

Normal Proton Background
NOAA/SWPC Boulder, CO USA

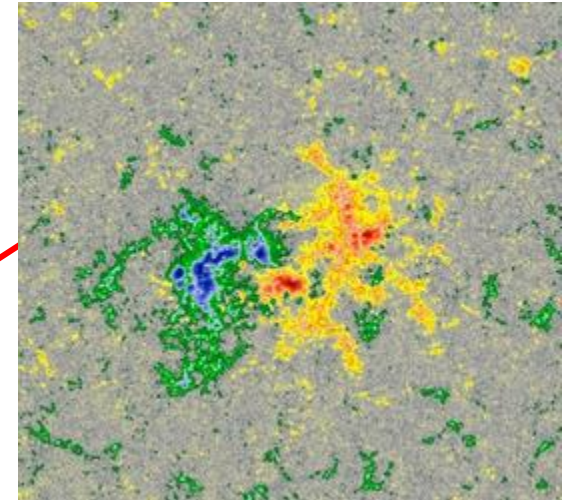
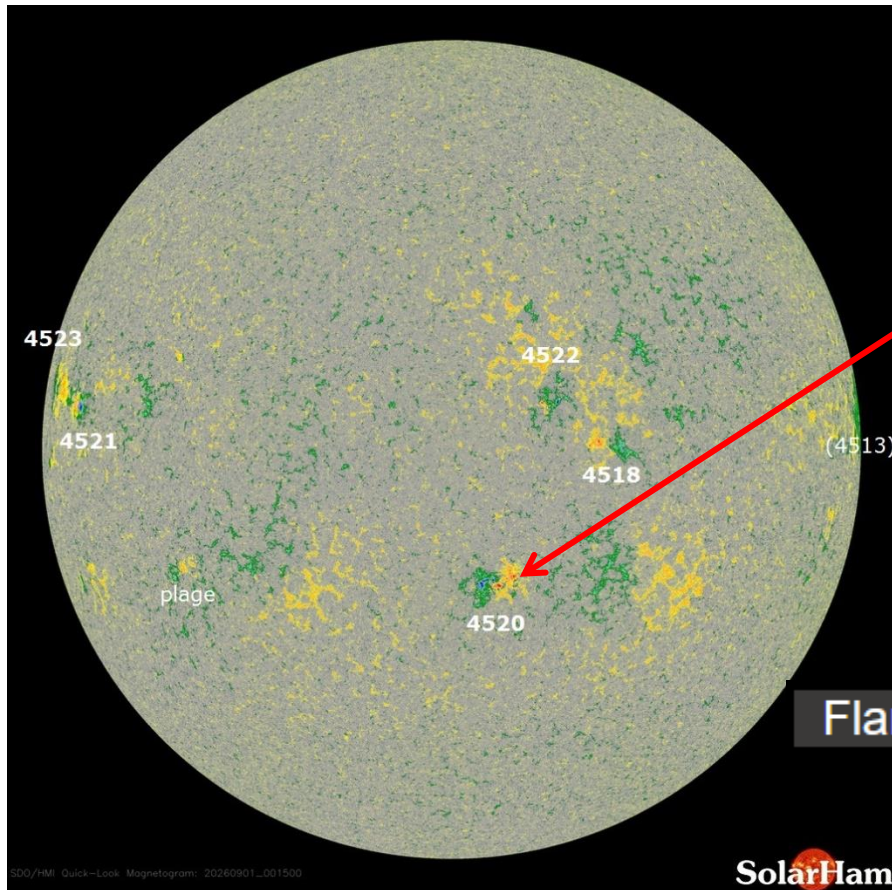
Global D-Layer Absorption + X-Ray Flux

Solar Demon

Solar Soft

Sun Spots

Magnetogram Image (Updated September 1, 2026)

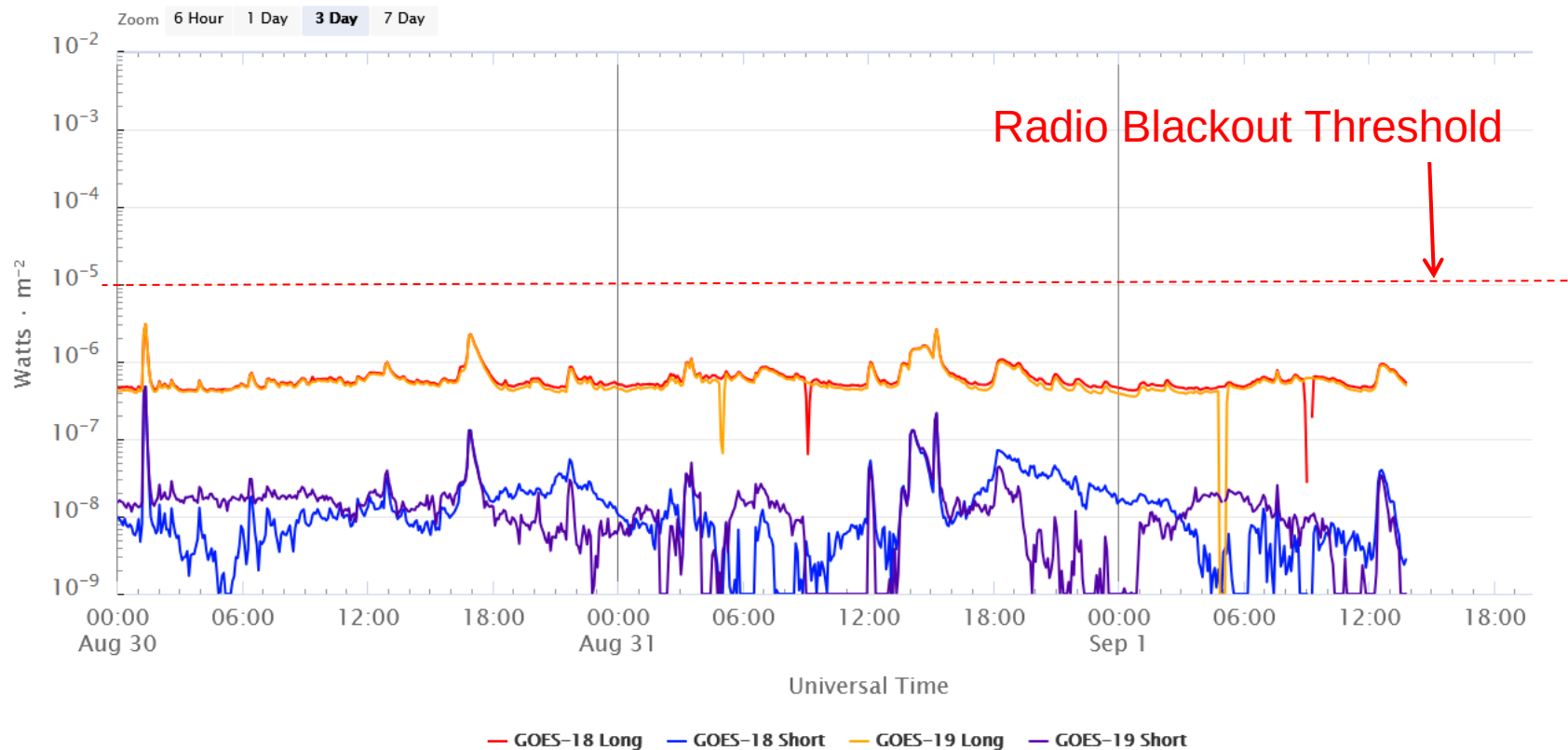


Beta

Flare Threat

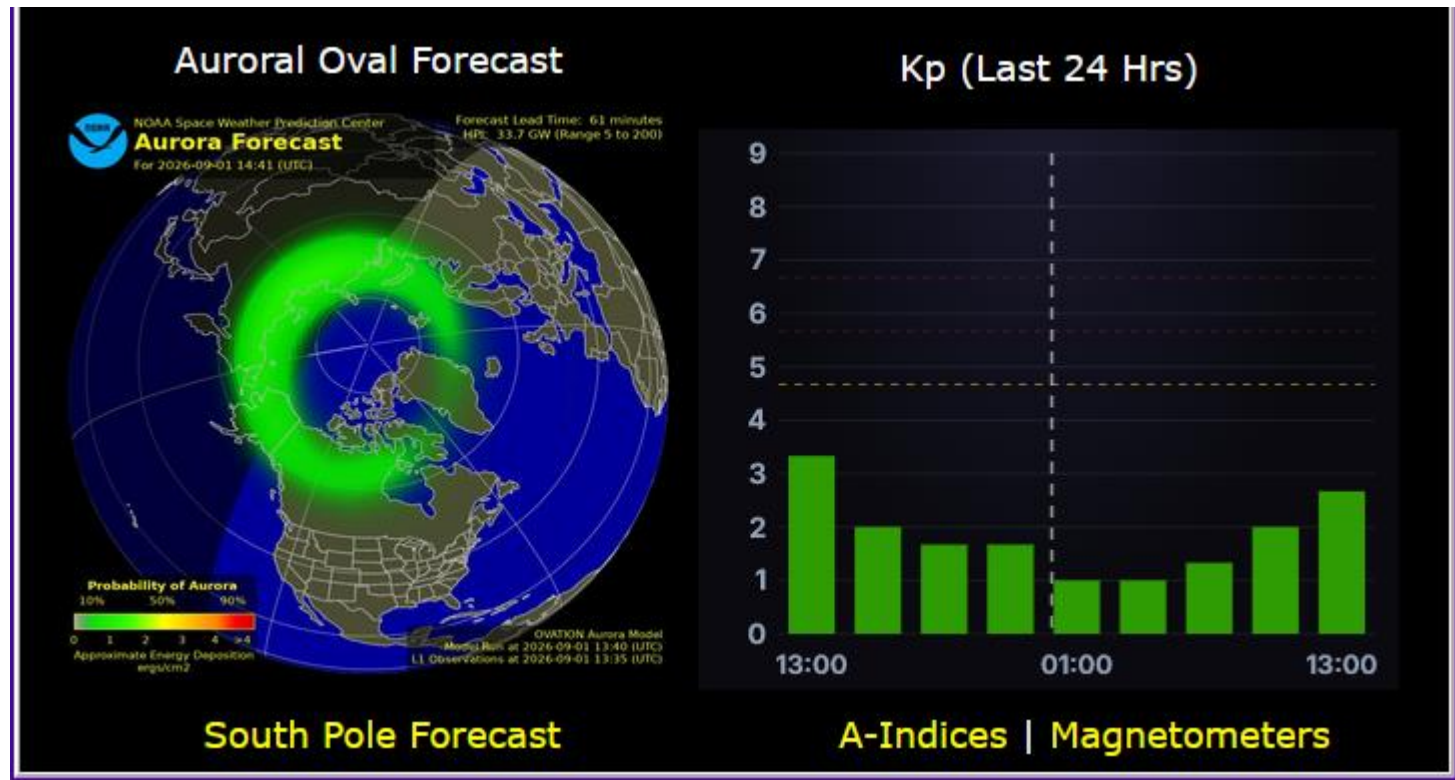
C: 50% **M: 10%** **X: 01%**

Solar X-Ray Flux: 30-31 AUG to 1 SEP 2026



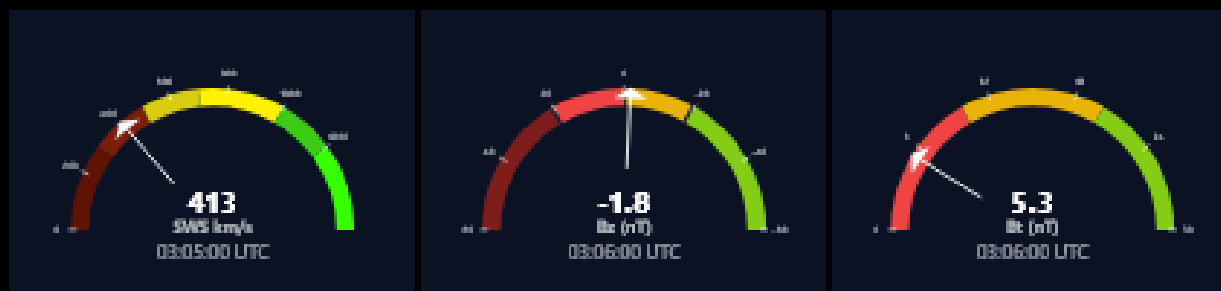
Earth's Geomagnetic Activity

1 SEP 2026



Real Time Solar Wind (Updated Every Minute)

SW Speed	Proton Density	Temperature	Bz / IMF	Bt (Strength)	Kp Index
413.1 km/s	3.58 cm ³	98028 K	-1.84 nT ↓	5.33 nT	1.33



Past 24 Hours Data Past 6 Hours



Geomagnetic Conditions: 1 SEP 2026

Solar wind:

$B_z = -1.8$ nT

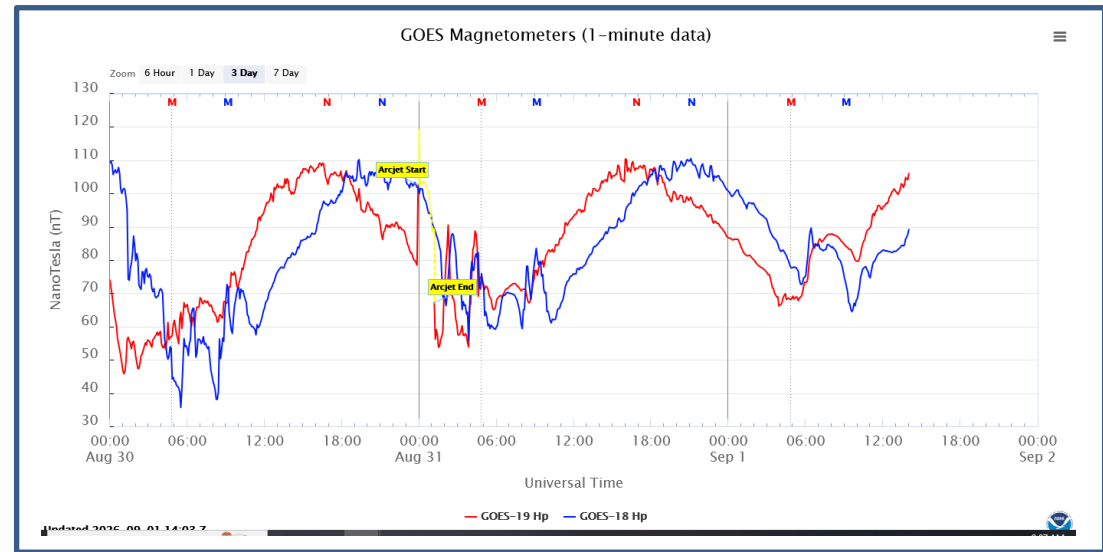
speed = 413 km/sec

density = 3.58 protons/cm³

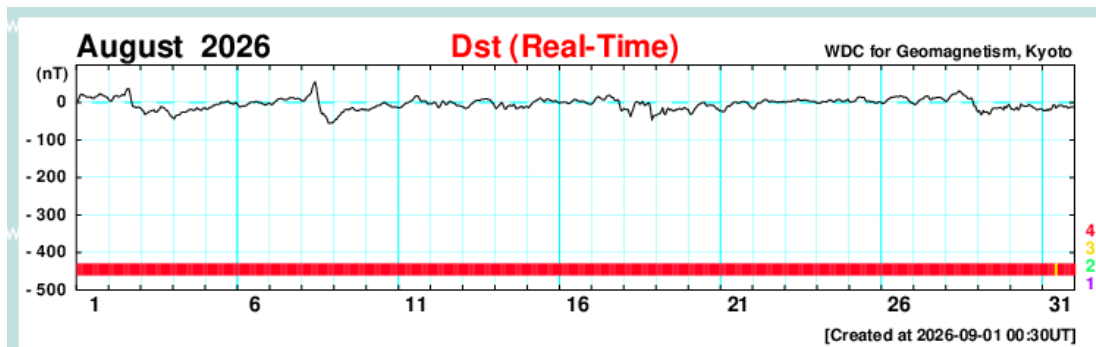
(From – NOAA DSCOVR
In L1, Lagrange Point)

Dst = 4 nT (Ring Field)

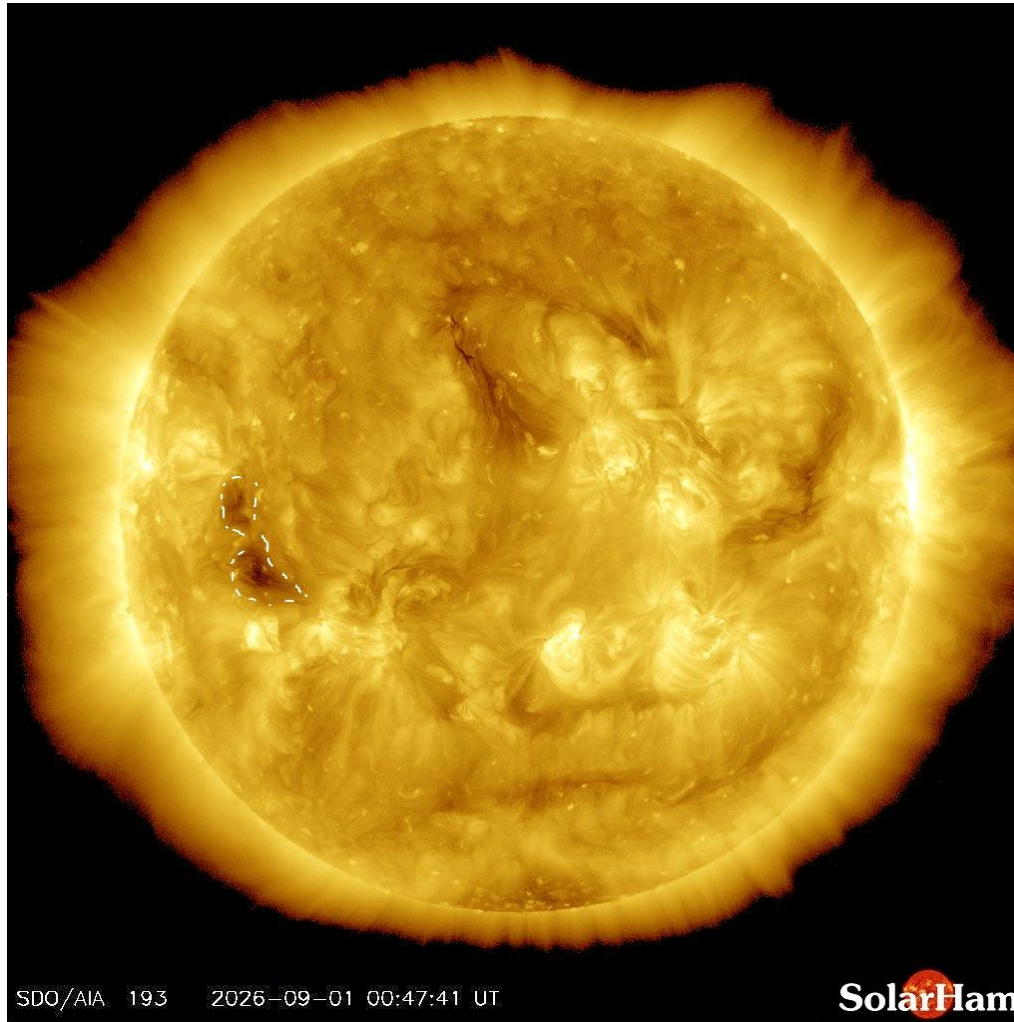
(From – Data Analysis Center
For Geomagnetism and Space
Magnetism – Kyoto University)



From – GOES 18, 19
In geostationary orbit



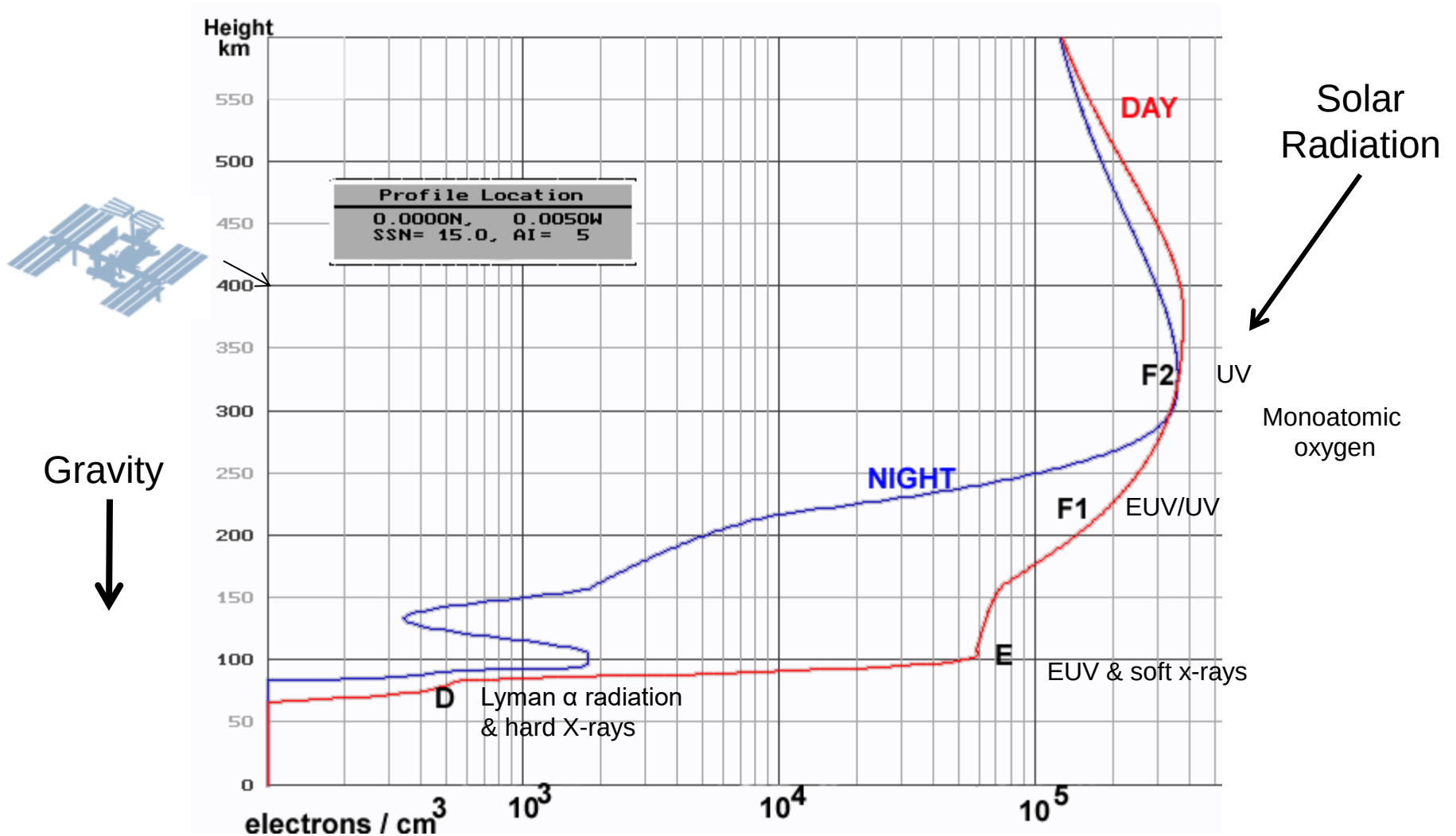
Coronal Holes – 1 SEP 2026

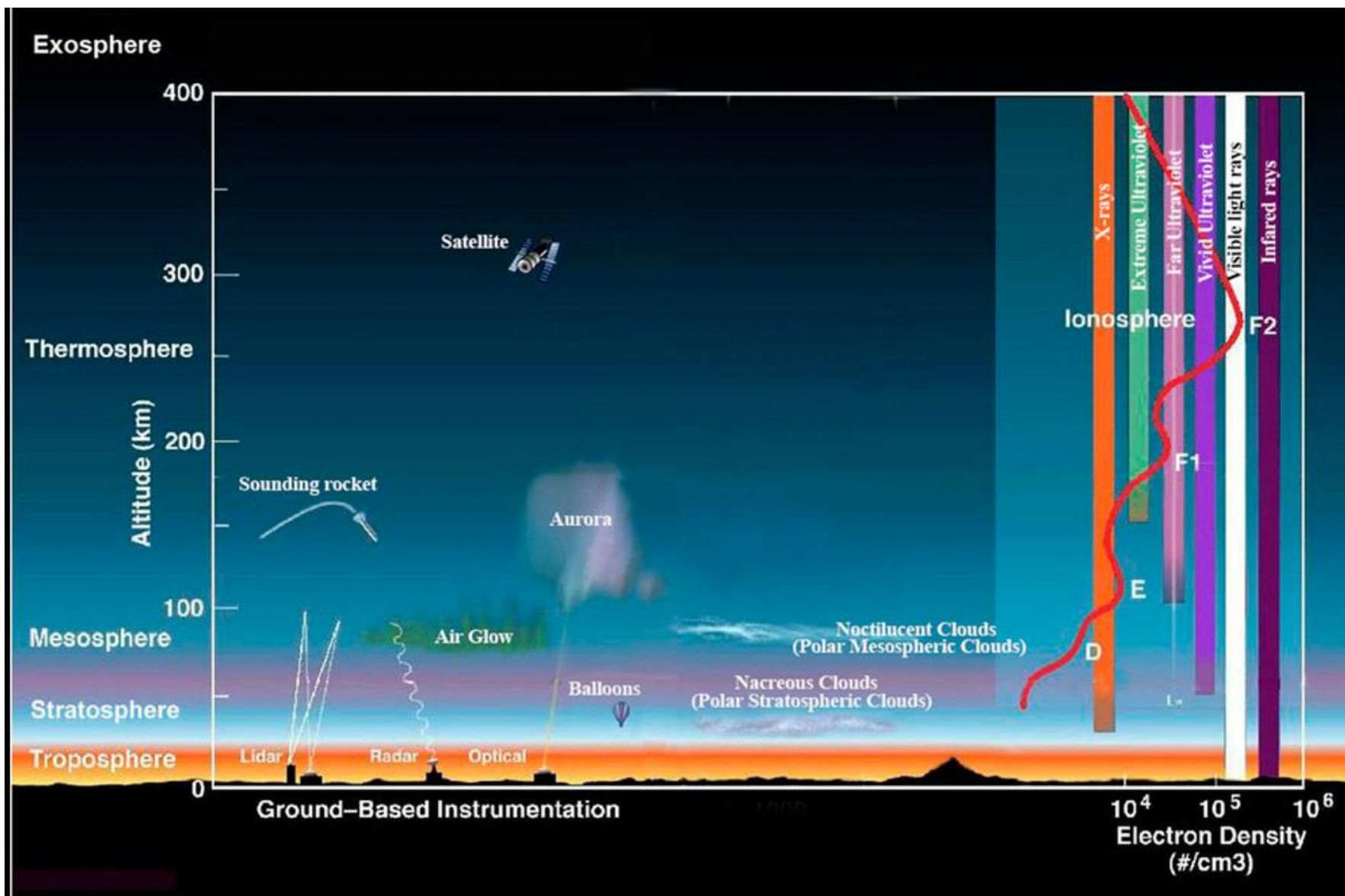


Analysis

There are currently no large coronal holes facing Earth.

Ionosphere Creation



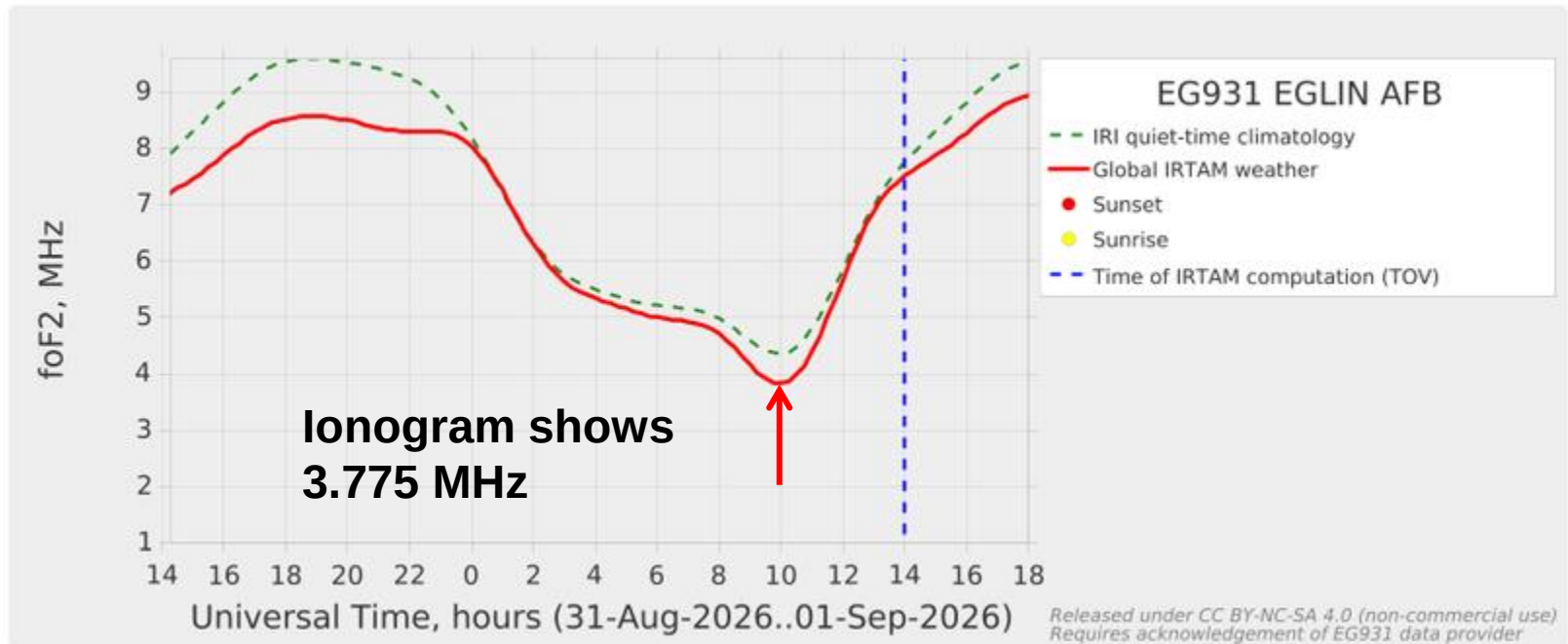


Austin Ionosonde Status

- GIRO is presently providing only GAMBIT models for both Austin and Eglin Ionosonde but these foF2 models appear to be accurate, based on MARS net performance (Critical Frequency).

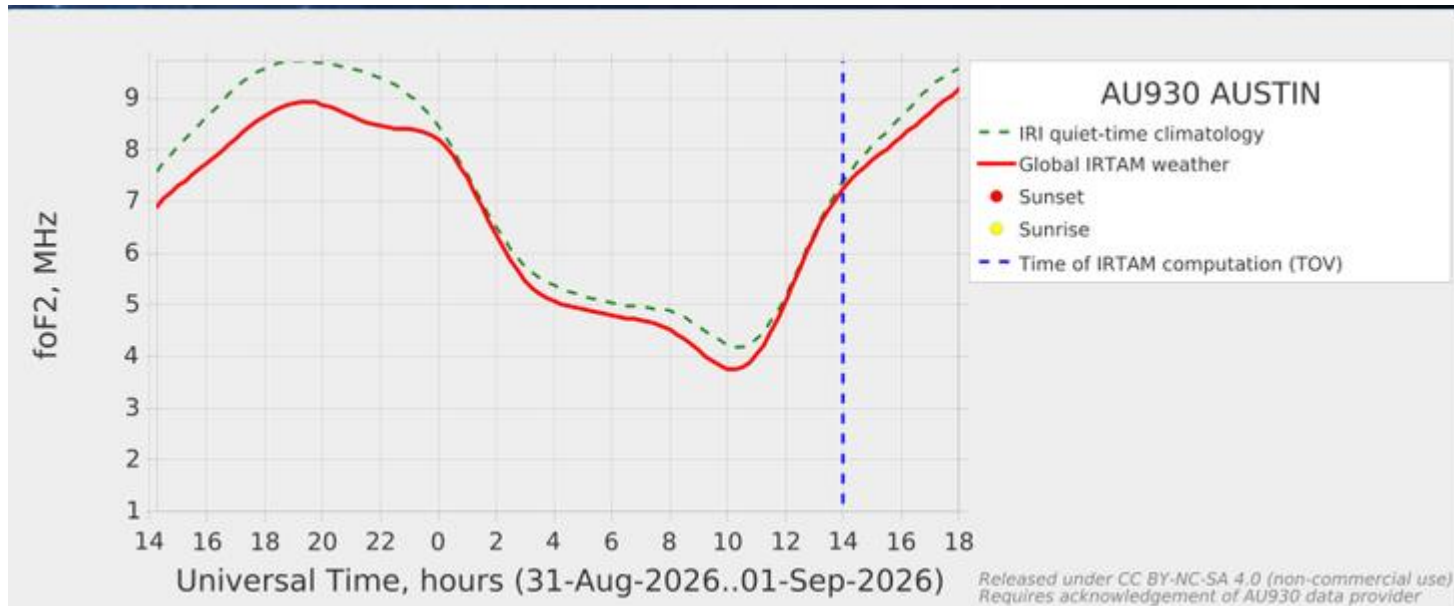
GAMBIT foF2 Trending Chart for Eglin Ionosonde

<https://www.region6armymars.org/resources/solarweather.php>



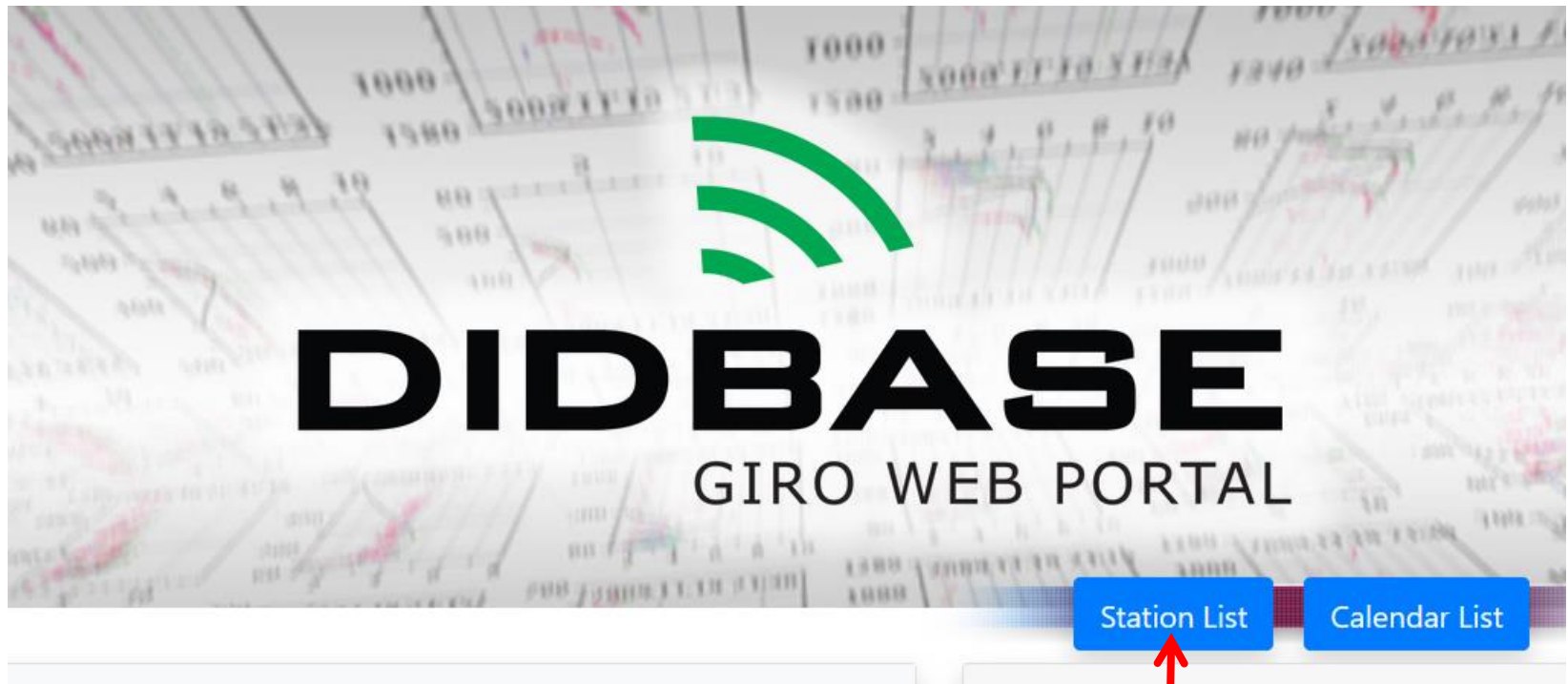
GAMBIT foF2 Trending Chart for Austin Ionosonde – Model only

<https://www.region6armymars.org/resources/solarweather.php>



Use of GIRO DIDBASE

<https://giro.uml.edu/didbase/>

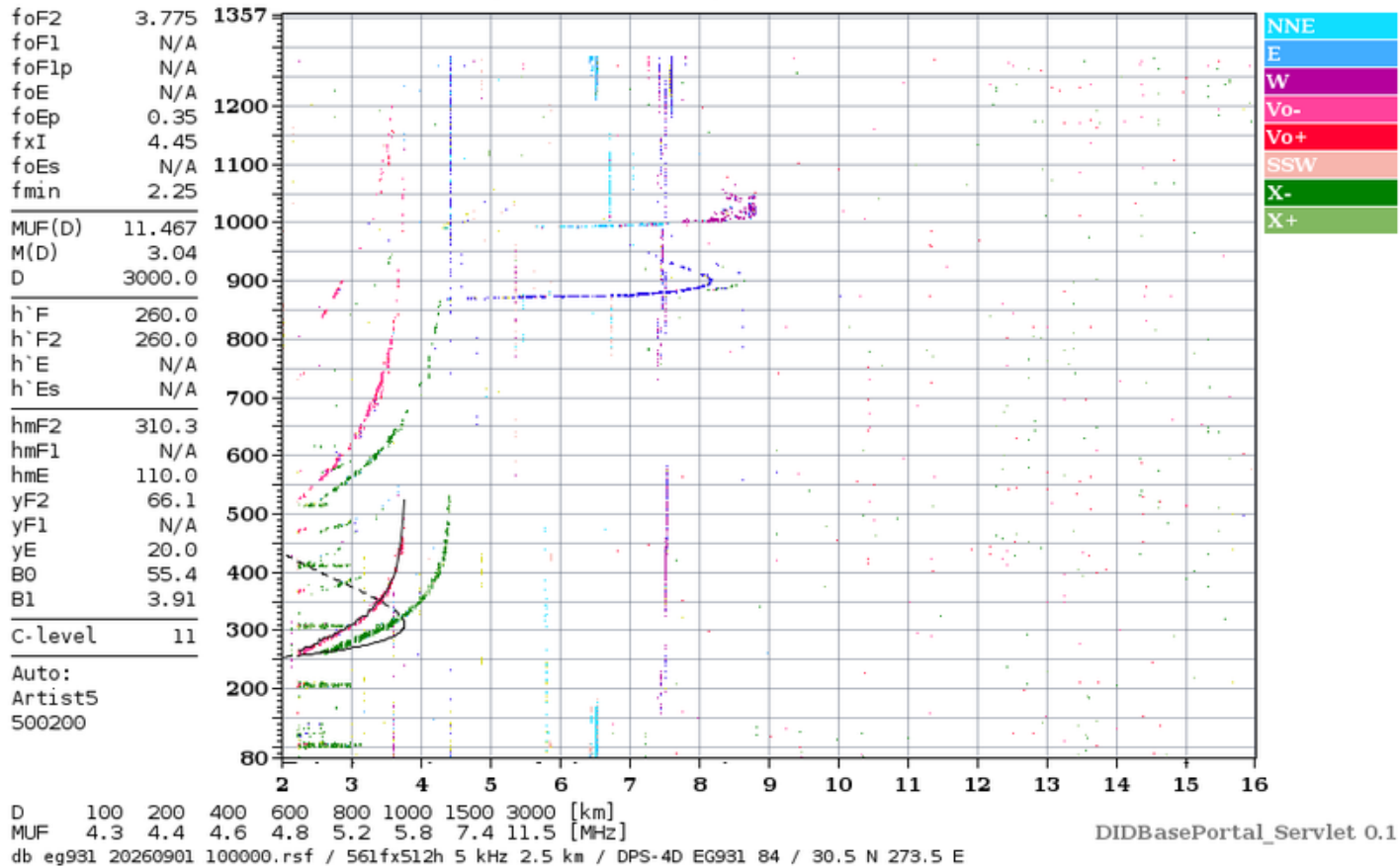


Eglin AFB

Eglin Ionosonde – 1000 DST

Lowell GIRO Data Center

Station YYYY DAY DDD HHMMSS P1 FFS S AXN PPS IGA PS
Eglin AFB 2026 Sep01 244 100000 RSF 1 712 100 03+ C0



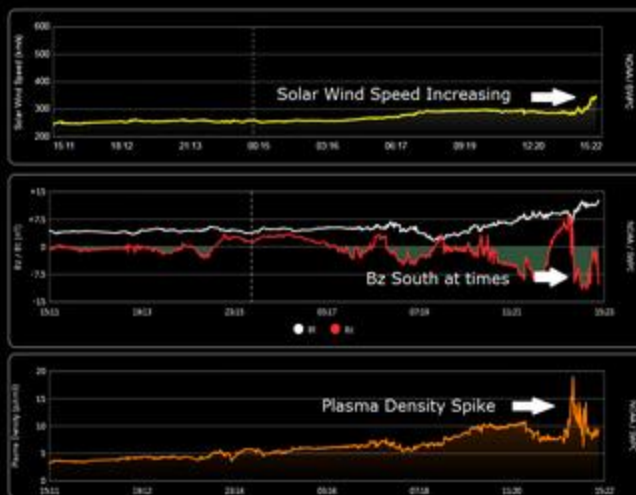
Time Shift for Eglin Ionosonde

- Eglin AFB Ionosonde is east of Austin at about the same Latitude so sun angle is about the same (same solar insolation).
- What happens to the Ionosphere over Eglin will occur over Austin about 45 minutes later.
- This is simply a Longitude difference converted to time.
- So look at Eglin's foF2 45 minutes earlier to see what is going to happen over Austin.
- The most rapid change in foF2 occurs as the sun is rising and setting (morning and evening).

Geomagnetic Update

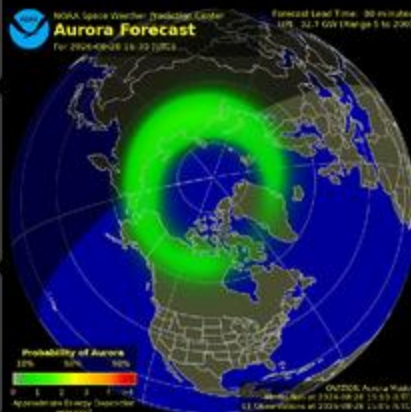
August 28, 2026 @ 15:20 UTC

The solar wind environment is now showing signs of activity as the speed increased further and the Bz/IMF component has been mostly pointing south. The active (Kp4) threshold was reached at 15:00 UTC (Aug 28). Minor (G1) geomagnetic storming may soon be possible. A watch for moderate (G2) conditions is still in effect for the next 24 hours due to possible incoming CME effects.



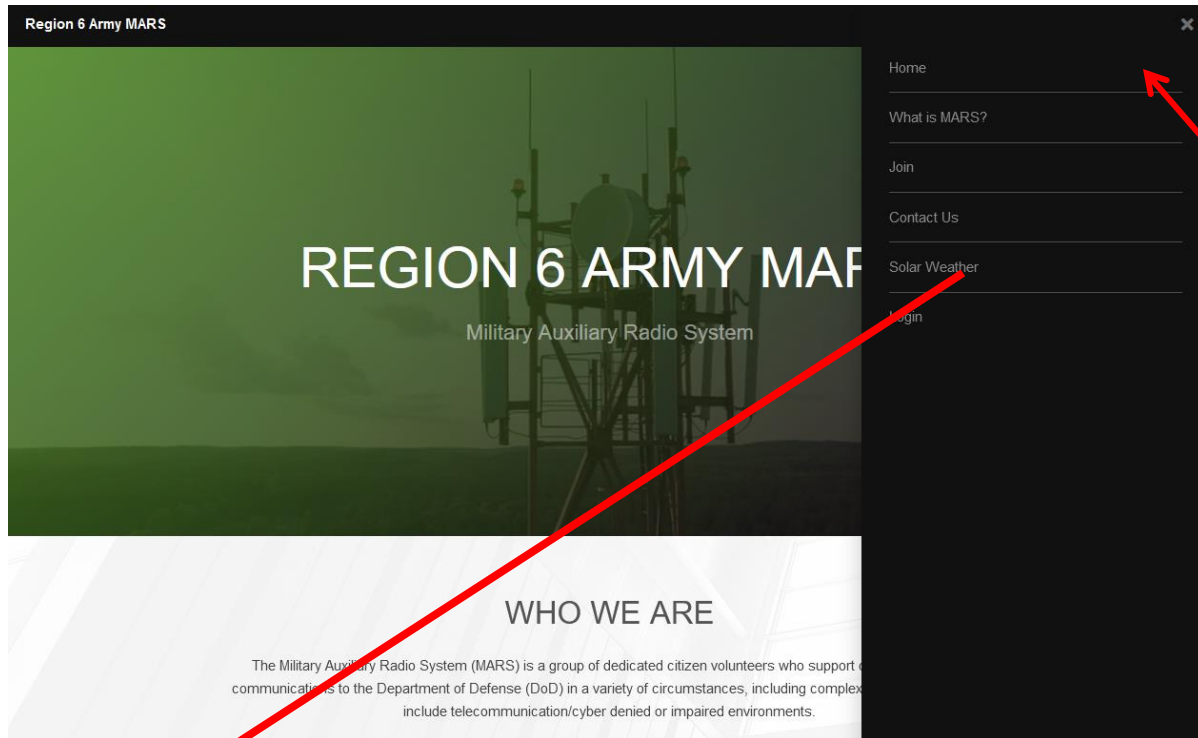
Kp 4 (Active) reached at 15:00 UTC (Aug 28)

Minor (G1) geomagnetic storming now possible



More News >>

Solar Weather Data



Menu

Solar Weather

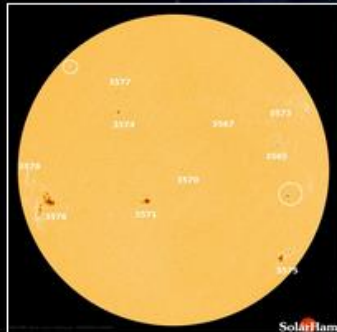
All Ionosondes
GAMBIT URL

- [GAMBIT](#) - Global Assimilative Model of Bottomside Ionosphere Timeline
 - [Boulder](#)
 - [Eglin](#)
- [NOAA Solar Weather](#) - Solar Weather plots of Kp and X-Ray and other solar emissions.
- [Solen Solar Weather](#) - Good general solar forecast from an individual.
- [Solar Ham](#) - SolarHam provides real time solar news, as well as consolidated data from various sources.

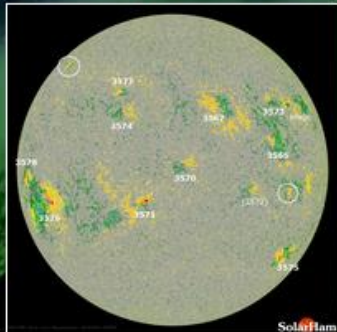
Space Weather for February 6, 2024

[Help Center + FAQ](#)

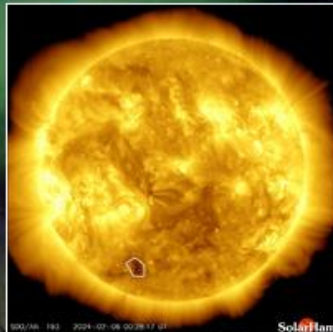
UTC Time 13:45:49 Tuesday



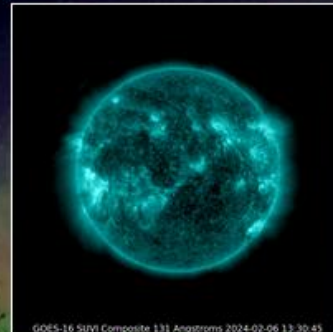
HMI Intensity
[Latest](#) | [Movie](#) | [HARP](#)



HMI Magnetogram
[Latest](#) | [Movie](#)



Coronal Holes
[Analysis](#) | [Movie](#)



SUVI 131 (Latest)
[Movie](#)



SUVI 304 (Latest)
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[Real Time Solar Wind](#)

[Protons and Electrons](#)

[Satellite Environment](#) ➔

Real Time Solar Wind (BETA) | [Expand Data](#)

Speed: 437 km/s

Density: 2.96 p/cm³

Bz: 4.25 nT ↑

Bt: 6.39 nT

Updated every minute.



<https://www.spaceweather.com/>

Current Conditions

Solar wind

speed: **314.8** km/sec

density: **9.9** protons/cm³

more data: [ACE](#), [DSCOVR](#)

Updated: Today at 1225 UT

X-ray Solar Flares

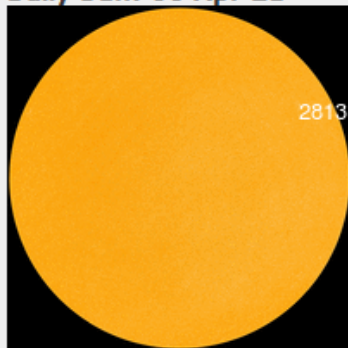
6-hr max: **A1** 1027 UT Apr06

24-hr: **A1** 1515 UT Apr05

[explanation](#) | [more data](#)

Updated: Today at: 1230 UT

Daily Sun: 06 Apr 21



Sunspot AR2813 is decaying, and poses no threat for strong flares.
Credit: SDO/HMI

FLYING TO THE VOLCANO: Iceland's Geldingadalur volcano has turned into an popular tourist attraction—especially since auroras were sighted [above the glowing lava](#). Early this morning, Tuesday, April 6th, Brian Emfinger saw auroras before he even reached the Reykjanes peninsula:



QUESTIONS?

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